

Divide Mixed Numbers

Lesson 2-4

Name:

Date:

Class:

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A mixed number consisting of the whole number 2 and the fraction 1/3. The 2 is in green, and the 1/3 is in red.

whole + fraction

2 1/3 means 2 wholes and 1/3 more — picture 2 full circles and 1/3 of another

Mixed Number

Write the definition:

An improper fraction with the numerator 11 and the denominator 4. The 11 is in red, and the 4 is in blue. The text 'top bigger' is written in green above the fraction line.

improper fraction

7/3 is improper because 7 > 3. It equals 2 1/3 as a mixed number.

Improper Fraction

Write the definition:



units of length

2 1/3 → multiply 2 × 3 = 6, add 1 → 7/3. Same value, different form.

Convert

Write the definition:



3/4

8/6: GCF is 2 → 8 ÷ 2 / 6 ÷ 2 = 4/3 = 1 1/3

Simplify

Write the definition:

$$\frac{3}{4} \rightarrow \frac{4}{3}$$

flip top and bottom

*The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$. To divide by $\frac{3}{4}$,
multiply by $\frac{4}{3}$.*

Reciprocal

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can divide mixed numbers by first changing them to improper fractions.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal

1 A number with a whole number and a fraction, like $2\frac{1}{3}$, is a _____.

2 A fraction whose numerator is greater than or equal to its denominator, like $\frac{7}{3}$, is an _____.

3 Changing a mixed number into an improper fraction is to _____ it.

4 Writing a fraction in its smallest equal form is to _____ it.

5 To divide by a fraction, I multiply by its _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is $1\frac{1}{2} \div \frac{3}{4}$?

- A. 2
- B. $1\frac{1}{8}$
- C. $\frac{3}{4}$
- D. $\frac{9}{8}$

Step 1 $1\frac{1}{2} = \frac{3}{2}$.

Step 2 Then $\frac{3}{2} \div \frac{3}{4} = \frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2$.

 **Answer:** A. 2

 We do — together

Solve this one with your class using the same steps.

Problem: Convert $3\frac{2}{5}$ to an improper fraction.

- A. $\frac{17}{5}$
- B. $\frac{15}{5}$
- C. $\frac{6}{5}$
- D. $\frac{32}{5}$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $2\frac{1}{4} \div \frac{3}{4}$?

- A. 3
- B. $1\frac{1}{2}$
- C. $\frac{9}{16}$
- D. $2\frac{1}{3}$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Mission 2: A surveillance route is 3 miles long, split into $1\frac{1}{2}$ -mile patrol sections. How many sections are on the route?

- A. 2
- B. $1\frac{1}{2}$
- C. $4\frac{1}{2}$
- D. 3

Show your work:

2. Mission 3: There are $7\frac{1}{2}$ feet of caution tape to seal a crime scene. Each doorway needs $2\frac{1}{2}$ feet. How many doorways can be sealed?

- A. 3
- B. 5
- C. $2\frac{1}{2}$
- D. 10

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective has a rope that is $5 \frac{1}{4}$ feet long. She needs to cut it into pieces that are each $\frac{3}{4}$ foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, keep-change-flip, multiply, simplify), and verify your answer.

Sentence starter: First I convert $5 \frac{1}{4}$ to _____. Then I use KCF: _____ $\times$ _____ = _____. The answer is _____ pieces. I can verify because _____ $\times \frac{3}{4} = 5 \frac{1}{4}$.

Show your work:

Reflect — Exit Ticket

What is $2 \frac{2}{3} \div \frac{2}{3}$?

- A. 4
- B. $\frac{2}{3}$
- C. $\frac{8}{3}$
- D. $1 \frac{7}{9}$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Mixed Number
2. **Notes 2:** Improper Fraction
3. **Notes 3:** Convert
4. **Notes 4:** Simplify
5. **Notes 5:** Reciprocal
6. **We Try (notes):** A. $17/5 - 3 \times 5 = 15$, then $15 + 2 = 17$. So $3 \frac{2}{5} = 17/5$.
7. **You Try (notes):** A. $3 - 2 \frac{1}{4} = 9/4$. Then $9/4 \div 3/4 = 9/4 \times 4/3 = 36/12 = 3$.
8. **Try It 1:** A. $2 - 3 \div 1 \frac{1}{2} = 3 \div 3/2 = 3 \times 2/3 = 6/3 = 2$ sections.
9. **Try It 2:** A. $3 - 7 \frac{1}{2} = 15/2$ and $2 \frac{1}{2} = 5/2$. So $15/2 \div 5/2 = 15/2 \times 2/5 = 30/10 = 3$ doorways.
10. **Exit Ticket:** A. $4 - 2 \frac{2}{3} = 8/3$. Then $8/3 \div 2/3 = 8/3 \times 3/2 = 24/6 = 4$.